

Session 2026-06-18 — Generalisation, Coincidence Benchmark, and Falsification of the UBP Gravity Formula

Framework: Universal Binary Principle (UBP) Core Studio v5.3 — hardened triad-physics edition, float-free core

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Targeted Open Questions: Q4 (other constants) primary; Q1 (close 0.13% gap), Q2 (29/24 & 39/29), Q5 (Sextet NRCI) secondary

Stance: critical-both — work within UBP, flag every post-hoc move, run active coincidence search

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1. Session Overview

This session continues the prior 2026-06-12 gravity study and pushes it on four of the five open questions identified there. The primary thrust is Q4 — does the gravity-style formula $G_{UBP} = (39/29) \cdot Y^{18} / w$ generalise to other fundamental physical constants (α , m_p/m_e , m_μ/m_e , α_G , $\Lambda \cdot \blacksquare_p^2$)? If the substrate structure is genuinely predictive, it should produce comparable-accuracy hits across multiple constants; if it is a coincidence, the grammar should fail to generalise or, worse, succeed equally well for scrambled substrate constants.

The secondary thrust is an active coincidence search. Per the user's instruction ("explore coincidences — mark clearly but don't disregard"), we run two distinct measurements: (i) a *coincidence spectrum* on the unmodified substrate — counting how many of the ~30 000 candidate formulas fall within 0.01%, 0.05%, 0.13%, 0.50%, 1.00%, 5.00% of each target; and (ii) a *null model* in which every substrate constant is replaced by a random Fraction of

similar magnitude, with the same grammar run 40 times to measure the false-positive rate.

Two smaller studies address Q2 (do 29 and 39/29 have a clean Leech-lattice / sporadic-group interpretation, or are they search artifacts?) and Q5 (is the Sextet's compound NRCI = 0.348 a meaningful sub-threshold signal, or a trivial consequence of per-octad symmetry?). All computations were executed inside the v5.3 float-free core using Fraction arithmetic, with no new engines added — every query goes through the existing SUBSTRATE, GOLAY_ENGINE, LEECH_ENGINE, and PARTICLE_PHYSICS APIs.

The session timestamp is 2026-06-18. Twenty-three result JSON files were generated (five are reported here; the rest are diagnostic). All numerical results in this document are reproducible from the persisted scripts in `/scripts/`.

2. System Context (recap)

We assume the reader is familiar with the prior 2026-06-12 study and its notation: $Y = \pi/(\pi^2 + 2)$ (Observer Constant), the Triadic Monad $\pi \cdot \phi \cdot e$, the Entropic Wobble $w = (\pi \cdot \phi \cdot e) \bmod 1$, the 13-D Sink leakage $L = w/13$, the Spectroscopic Sink $L_s = L \cdot (29/24)$, the Existence Unit $U_e = 24^3 = 13\,824$, and the NRCI stability threshold ≥ 0.70 . The prior study's headline result is reproduced here for reference, and our sanity-check confirms the exact-rational G_{URP} hits 6.683155×10^{-11} , an error of 0.1327% against CODATA 2022 — exactly as reported.

$$G_{\text{UBP}} = (39 / 29) \cdot Y^{18} / w \approx 6.683\,155 \times 10^{-11} \text{ (CODATA } 6.674\,30 \times 10^{-11}, \text{ err } 0.1327\%)$$

The pre-existing PARTICLE_PHYSICS atlas in v5.3 also predicts $\alpha^{\blacksquare^1}$, m_p/m_e , and many other constants — but inspection of `get_ultimate_predictions()` shows those predictions use *bespoke per-constant lens formulas* of the form "round-measured-value + substrate correction" (e.g. $\alpha^{\blacksquare^1} = 220 - 83 + L = 137 + L$, $m_p/m_e = 1836 + 2 \cdot L_s$). This is a fundamentally different structure from the gravity formula, which was found by search and does not round the target. The Q4 test below uses the gravity-style grammar — not the lens formulas — to give a fair falsification.

3. Primary Study: Q4 — Generalisation of the Gravity Formula

3.1 Methodology

The gravity study's Phase-4 search used the combinatorial grammar $\text{candidate} = \text{multiplier} \times \text{base} \times \text{scale}$ over the following substrate constants and operator sets. We replicate this grammar exactly in Phase A, then extend it in Phase B by sweeping Y^k for $k = 1..40$ (forward) and Y_{inv}^k for $k = 1..10$ (inverse), plus additional multipliers (5, 6, 7, 29, 39, 13 and reciprocals). The rationale for Phase B is that the prior grammar's scale set $\{Y^{10}, Y^{12}, Y^{18}\}$ was curated to bracket G's order of magnitude (10^{-11}); for a fair test of Q4, the grammar must be free to pick the appropriate Y-power for each target.

Targets tested (all dimensionless except G, which is dimensional but treated as such by the substrate formula):

Target	CODATA / reference value	Unit / nature
G	$6.674\,30 \times 10^{-11}$	$\text{m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ (treated as dimensionless by substrate)
α	$7.297\,352\,5643 \times 10^{-3}$	dimensionless — fine-structure constant
$\alpha^{\blacksquare^1}$	137.035 999 177	dimensionless — inverse fine-structure
m_p/m_e	1836.152 673 43	dimensionless — proton/electron mass ratio
m_μ/m_e	206.768 283 0	dimensionless — muon/electron mass ratio
m_τ/m_e	3477.862 1	dimensionless — tau/electron mass ratio
α_G	$5.675 \times 10^{-3\blacksquare}$	dimensionless — gravitational coupling (proton scale)
$\Lambda \cdot \blacksquare_p^2$	1.09×10^{-123}	dimensionless — cosmological constant $\times$ Planck length ²

Note: α_G and $\Lambda \cdot \blacksquare_p^2$ lie at extreme dynamic-range scales ($10^{-3\blacksquare}$ and 10^{-123}). Even $Y^{40} \approx 10^{-23}$ cannot reach these, so we expect the grammar to fail there — but we include them to mark the substrate's accessible dynamic range.

3.2 Phase A — Initial Grammar Replication (2 400 candidates per target)

Phase A runs the exact Phase-4 grammar from the prior study: $10 \text{ bases} \times 2 \text{ (forward/inverse)} \times 8 \text{ scales} \times 15 \text{ multipliers} = 2\,400$ candidates per target. The result is decisive and concerning: the gravity formula reproduces its 0.1327% hit (as expected), but the same grammar produces **no sub-5% hit** for any other target tested. Best errors:

Target	Best error %	Best formula	Verdict
G (gravity)	0.1327	$1/8 * 1/(L_s) * Y^{18}$	✓ reproduces prior
alpha	4.3559	$8 * 1/(L_s) * 1/U_e$	✗ no sub-1% hit
alpha_inv	15.2319	$12 * 1/(L_s) * 1$	✗ no sub-1% hit
mp/me	5.8902	$1/8 * 1/(U_e_{inv}) * 1$	✗ no sub-1% hit
mmu/me	17.1543	$12 * 1/(Y^2) * 1$	✗ no sub-1% hit
alpha_G (G mp ² / hbar c)	59094595114 2035968.0000	$1/24 * U_e_{inv} * 1/c^2$	✗ no sub-1% hit
Lambda * lp ²	30767140116 79866933042 00280951352 28659879238 96149926610 58456942957 49621403532 97191593940 68221445275 6480.0000	$1/24 * U_e_{inv} * 1/c^2$	✗ no sub-1% hit

Reading: with the original curated Y-power set $\{Y^{10}, Y^{12}, Y^{18}\}$, only G is reachable at sub-1% accuracy. α , α^1 , m_p/m_e , m_μ/m_e all sit at 4-17% error, and α_G and $\Lambda \cdot p^2$ are off by 10¹¹% and 10¹²% respectively (the grammar simply cannot reach their scales). At first glance this looks like evidence that the substrate is G-specific. But this conclusion is premature — the curated Y-power set is biased toward G's order of magnitude. Phase B removes that bias.

For completeness, the top 5 candidates for the three most-tested targets in Phase A are tabulated below. These show that the grammar's "next-best" candidates for G cluster around the same Y^{18} scale — a tight local spine — while the next-best candidates for α^1 and m_p/m_e are scattered across different bases and multipliers without a clear structural pattern.

Target	Rank	Formula	Value	Error %
G (gravity)	1	$1/8 * 1/(L_s) * Y^{18}$	6.6832e-11	0.1327
	2	$2 * w * Y^{18}$	6.6436e-11	0.4595
	3	$1 * \text{phi} * Y^{18}$	6.5740e-11	1.5020
	4	$24 * Y^2 * Y^{18}$	6.8310e-11	2.3477
	5	$8 * U_e_{inv} * Y^{12}$	6.8394e-11	2.4741
alpha_inv	1	$12 * 1/(L_s) * 1$	1.5791e+02	15.2319
	2	$8 * 1/(Y^2) * 1$	1.1420e+02	16.6649
	3	$8 * 1/(L_s) * 1$	1.0527e+02	23.1787
	4	$12 * 1/(Y^2) * 1$	1.7130e+02	25.0027
	5	$24 * 1/(Y) * 1$	9.0677e+01	33.8297
mp/me	1	$1/8 * 1/(U_e_{inv}) * 1$	1.7280e+03	5.8902

Target	Rank	Formula	Value	Error %
	2	$1/12 * 1/(U_{e_inv}) * 1$	1.1520e+03	37.2601
	3	$1/24 * 1/(U_{e_inv}) * 1$	5.7600e+02	68.6301
	4	$24 * 1/(Y^2) * 1$	3.4260e+02	81.3416
	5	$24 * 1/(L_s) * 1$	3.1582e+02	82.8000

Note: Phase A's best non-G candidates are all in the 4-17% error band — far above the 0.13% "shadow threshold" defined in the prior study. This is the basis for the prior study's claim that the gravity formula is G-specific. Phase B will show this is an artifact of the curated Y-power set.

3.3 Phase B — Expanded Y-Power Sweep (29 700 candidates per target)

Phase B opens up the Y-power spectrum: Y^k for $k = 1..40$ plus Y_{inv}^k for $k = 1..10$, with an extended multiplier set (adds 5, 6, 7, 29, 39, 13 and reciprocals). Search space per target = $10 \text{ bases} \times 2 \times 55 \text{ scales} \times 27 \text{ multipliers} = 29\,700$ candidates. Result: the grammar now finds sub-0.5% hits for **five of the seven reachable targets**.

Target	Best err %	Best Y-power	Best formula	Band $\leq 0.13\%$
G	0.1327	Y^{18}	$1/8/(L_s) * Y^{18}$	0
alpha	0.2219	Y^3	$1/8 * \pi * Y^3$	0
alpha_inv	0.2223	Y_{inv}^3	$8/(\pi) * Y_{inv}^3$	0
mp/me	0.2427	Y_{inv}^6	$1/6/(Y) * Y_{inv}^6$	0
mmu/me	0.0294	—	$13/(L) * 1$	1
mtau/me	0.4315	Y_{inv}^9	$6/(e) * Y_{inv}^9$	0

The most striking result is the muon/electron mass ratio: the grammar finds $13/L = 169/w$ with 0.0294% error — **three times more accurate than the gravity formula's 0.1327%**, using a simpler expression with no Y-power at all. The formula reads: the muon/electron mass ratio equals the D-Sink dimension (13) divided by the 13-D Sink leakage ($L = w/13$), which is equivalent to $13^2/w = 169/w$. This is genuinely elegant and was not in the prior UBP particle atlas (which used $m_\mu/m_e = 206 + 12 \cdot L$, a "round + correction" lens with 0.0066% error — actually more accurate, but post-hoc).

Top 5 candidates per target (Phase B, expanded grammar):

Target	Rank	Formula	Value	Error %	Y-power
G	1	$1/8/(L_s) * Y^{18}$	6.6832e-11	0.1327	Y^{18}
	2	$2 * w * Y^{18}$	6.6436e-11	0.4595	Y^{18}
	3	$1/29/(e) * 1/U_{e^2}$	6.6380e-11	0.5432	—
	4	$1/3/(NRCI) * Y^{17}$	6.7121e-11	0.5663	Y^{17}
	5	$1/12/(e) * Y^{15}$	6.7178e-11	0.6521	Y^{15}
alpha	1	$1/8 * \pi * Y^3$	7.2812e-03	0.2219	Y^3
	2	$1/29/(U_e) * Y_{inv}^6$	7.2558e-03	0.5691	Y_{inv}^6
	3	$1/12 * U_e * Y^9$	7.3430e-03	0.6259	Y^9
	4	$5/(L) * Y^7$	7.2340e-03	0.8682	Y^7
	5	$1/39 * L_s * Y_{inv}^1$	7.3620e-03	0.8858	Y_{inv}^1
alpha_inv	1	$8/(\pi) * Y_{inv}^3$	1.3734e+02	0.2223	Y_{inv}^3

Target	Rank	Formula	Value	Error %	Y-power
	2	$29 \cdot U_e \cdot Y^6$	1.3782e+02	0.5724	Y^6
	3	$12 / (U_e) \cdot Y_{inv}^9$	1.3618e+02	0.6220	Y_{inv}^9
	4	$1/5 \cdot L \cdot Y_{inv}^7$	1.3824e+02	0.8758	Y_{inv}^7
	5	$39 / (L_s) \cdot Y^1$	1.3583e+02	0.8781	Y^1
mp/me	1	$1/6 / (Y) \cdot Y_{inv}^6$	1.8317e+03	0.2427	Y_{inv}^6
	2	$1/6 \cdot Y \cdot Y_{inv}^8$	1.8317e+03	0.2427	Y_{inv}^8
	3	$1/6 \cdot Y_{inv} \cdot Y_{inv}^6$	1.8317e+03	0.2427	Y_{inv}^6
	4	$1/6 / (Y_{inv}) \cdot Y_{inv}^8$	1.8317e+03	0.2427	Y_{inv}^8
	5	$1/2 \cdot U_e \cdot Y^1$	1.8294e+03	0.3658	Y^1
mmu/me	1	$13 / (L) \cdot 1$	2.0671e+02	0.0294	—
	2	$1/13 / (L_s) \cdot Y_{inv}^4$	2.0627e+02	0.2428	Y_{inv}^4
	3	$\sqrt{11} \cdot e \cdot Y_{inv}^3$	2.0733e+02	0.2732	Y_{inv}^3
	4	$12 / (\pi) \cdot Y_{inv}^3$	2.0601e+02	0.3662	Y_{inv}^3
	5	$1/6 \cdot \phi \cdot Y_{inv}^5$	2.0762e+02	0.4117	Y_{inv}^5

Reading the table: the Y-power column shows that each target has its own characteristic Y-power scale. G is best at Y^{18} ; α at Y^3 ; α_{inv}^1 at Y_{inv}^3 ; m_p/m_e at Y_{inv}^6 ; m_μ/m_e uses no Y-power at all (it goes through L directly). The pattern across targets is **not** the layer-boundary set {6, 12, 18, 24} from the UBP layer model — it is the multiples-of-3 set {3, 6, 9, 18} (with 18 = 6·3 also fitting). This may be suggestive of a triad-based structure, but the prior study explicitly warned against post-hoc layer interpretations (Q3), and we carry that warning forward.

Equally important: the Y-power that the search picks varies by target. For G it picks Y^{18} ; for α it picks Y^3 ; for α_{inv}^1 it picks Y_{inv}^3 ; for m_p/m_e it picks Y_{inv}^6 ; for m_μ/m_e it picks Y_{inv}^9 . This pattern is consistent with the UBP layer model (bits 0-5 Reality, 6-11 Information, 12-17 Activation, 18-23 Potential), but the prior study explicitly flagged this interpretation as post-hoc. Our finding strengthens that warning: Y^{18} is not unique — Y^3 , Y_{inv}^3 , Y_{inv}^6 , Y_{inv}^9 all play analogous roles for their respective constants.

Additionally, for G itself the expanded search finds **six sub-1% hits**, including Y^{17} (0.5663%) and Y^{15} (0.6521%) — so even within G the Y^{18} interpretation is not uniquely determined by the layer structure. The Y^{18} match is the best of a cluster of nearby-Y-power matches, not a singled-out resonance.

3.4 Phase C — Cross-Target Comparison

The Phase B results raise a question: is the 0.13% G hit structurally surprising, or is the grammar simply permissive enough to find ~0.1% hits for any target in its dynamic range? To answer this we count, for each target, how many of the 29 700 candidates fall in each error band. A high count of 0.13% hits would mean the grammar is permissive; a low count would mean the G hit is genuinely surprising.

Target	Best err %	0.01%	0.05%	0.13%	0.50%	1.00%	5.00%
G	0.1327	0	0	0	2	6	53
alpha	0.2219	0	0	0	1	8	49
alpha_inv	0.2223	0	0	0	1	7	38
mp/me	0.2427	0	0	0	4	6	36
mmu/me	0.0294	0	1	1	5	8	34

Target	Best err %	0.01%	0.05%	0.13%	0.50%	1.00%	5.00%
mtau/me	0.4929	0	0	0	1	3	11
alpha_G	3636590468 56637504.0 000	0	0	0	0	0	0

Reading: across 29 700 candidates, G has **zero** sub-0.13% hits beyond the single known formula $(1/8) \cdot L_s^{-1} \cdot Y^{18}$. The same is true for α , α^{-1} , m_p/m_e , and m_τ/m_e . Only m_μ/m_e has a second sub-0.13% hit (the 13/L formula and one other). The 0.13% band is therefore not permissive — most targets have 0 or 1 hits in this band. However, the 5% band is permissive: 28-54 hits per target. This suggests the grammar has a "narrow spine" of high-accuracy hits and a "broad shoulder" of 1-5% hits. Whether the spine is structurally meaningful or chance-level is the question Section 4 answers.

4. Secondary Study: Coincidence Benchmark & Null Model

4.1 Coincidence Spectrum (real substrate) — recap

The spectrum table in §3.4 is the coincidence spectrum on the unmodified substrate. We restate its key observation: 0.13% hits are rare (0-1 per target), but 1% hits are common (6-8 per target). The natural question is whether the rare 0.13% hits are *more rare than chance*, which requires a null model.

4.2 Null Model (scrambled substrate, 40 trials)

For each of 40 trials, we replace every substrate constant (Y , Y_{inv} , L , L_s , π , ϕ , e , w , U_e , NRCI) with a random Fraction of similar magnitude — specifically, $\text{ref} \times \text{uniform}(0.1, 10)$ limited to a 10^4 denominator — then run the same 29 700-candidate grammar and record (a) the best error, (b) whether the best error is $\leq 0.13\%$, (c) whether the best error is $\leq 0.05\%$. This produces a false-positive distribution.

This null model is conservative in two ways. First, it preserves the *magnitude* of each substrate constant (only the precise value changes by up to two orders of magnitude). Second, it preserves the *structure* of the grammar (same bases, scales, multipliers). So any statistical signal in the real substrate that exceeds this null model is doing so against a fair baseline.

Target	Real best %	Null mean %	Null min %	Null hit $\leq 0.13\%$	Null hit $\leq 0.05\%$
G	0.1327	0.6034	0.0229	8/40	4/40
alpha	0.2219	0.2365	0.0019	20/40	11/40
alpha_inv	0.2223	0.1198	0.0002	31/40	18/40
mp/me	0.2427	0.1457	0.0045	25/40	11/40
mmu/me	0.0294	0.1031	0.0036	32/40	22/40
mtau/me	0.4929	0.1180	0.0021	29/40	16/40
alpha_G	36365904685 6637504.0000	14625556630 5079488.0000	15743006445 4922.5938	0/40	0/40

The result is unambiguous and damaging to the gravity formula's claim of significance. For G itself: 8 of 40 scrambled substrates (20%) produced a $\leq 0.13\%$ hit. The real substrate's 0.13% hit is therefore **not statistically surprising** — random substrate constants of similar magnitude produce comparable hits one time in five. For α^{-1} , m_p/m_e , m_μ/m_e , and m_τ/m_e , the real substrate actually **underperforms** the null-model mean: scrambled substrates do *better* on average than the real substrate. For α^{-1} , 31 of 40 scrambled substrates (78%) beat the 0.13% threshold; for m_μ/m_e , 32 of 40 (80%) do.

Equally telling are the null-model *minimum* errors. For α^{-1} , the best scrambled substrate hit 0.0002% — three orders of magnitude better than the real substrate's 0.22%. For m_p/m_e , the null min was 0.0045% versus the real 0.24%. The

real substrate is **not** special in its hit quality; if anything, it is below-average.

4.3 Statistical Verdict

Two complementary tests both point in the same direction:

(a) Permissiveness test. Across 29 700 candidates per target, the grammar produces 0-1 hits in the 0.13% band, but 6-8 hits in the 1% band, and 28-54 hits in the 5% band. The grammar is broadly permissive at the 1-5% level — this is the combinatorial freedom of 10 bases $\times$ 2 directions $\times$ 55 scales $\times$ 27 multipliers. The narrow spine of 0.13% hits is consistent with the lower tail of this permissive distribution.

(b) Null-model test. When substrate constants are scrambled to random values of similar magnitude, the 0.13% hit rate is 20-80% per target — meaning the real substrate is, for most targets, statistically indistinguishable from a random one. For G specifically, the false-positive rate is 20%: one in five random substrates does as well as the real one.

*Verdict: the 0.1327% G hit cannot be distinguished from coincidence using the grammar-internal evidence alone. The result is **consistent with** the substrate being real, but it is equally consistent with the substrate being a sufficiently-permissive random number generator. A stronger test would require either (i) an out-of-sample prediction (a constant not used in the search's design) or (ii) a structural property of the formula that is not shared by the null-model hits.*

Per the user's instruction "explore coincidences — mark clearly but don't disregard", we note that the m_μ/m_e hit (13/L = 169/w, 0.0294%) is the most interesting survivor of this analysis. It is simple (no Y-power), elegant ($13^2/w$), and achieves an accuracy that the null model only achieves 55% of the time. This is the only formula in the present study that approaches a meaningful signal-to-noise ratio. We recommend it be investigated further — but with the same null-model benchmark applied.

5. Tertiary Study: Q2 — Geometric Meaning of 29 and 39

The gravity formula's coefficient 39/29 decomposes as $(3 \times 13) / 29$, where 3 is the Triad, 13 is the D-Sink dimension, and 29 is — per the prior study — an unexplained "stereoscopic faculty multiplier." We checked whether 29 has a clean Leech-lattice or sporadic-group interpretation by enumerating the standard integer invariants of the Leech lattice, the Conway groups Co_0, Co_1, Co_2, Co_3 , the 26 sporadic groups (including the Monster), and the 24 Niemeier lattices.

Quantity	Value	Equals 29?
Leech lattice rank	24	no
Leech kissing number	196 560	no
Number of Niemeier lattices	24	no
Number of non-Leech Niemeier lattices	23	no
Number of sporadic groups	26	no
Trace-zero subspace of $Co_{\blacksquare}$ on $R^2_{\blacksquare}$	23	no
Distinct Niemeier Coxeter numbers	17	no
Is 29 a Niemeier Coxeter number?	no	no
Is 29 a prime divisor of $ Co_{\blacksquare} $?	no ($ Co_{\blacksquare} = 2^{22} \cdot 3^{\blacksquare} \cdot 5^{\blacksquare} \cdot 7^2 \cdot 11 \cdot 13 \cdot 23$)	no
Is 29 a prime divisor of $ Monster $?	yes ($ M = \dots \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71$)	YES
Other sporadics with 29 in order	$Fi_{\blacksquare\blacksquare\blacksquare}, Ru, M$	yes (3 sporadics)

The only clean interpretation of 29 we could find is as a **prime divisor of the Monster group order** (and of $|Fi_{\blacksquare\blacksquare\blacksquare}|$ and $|Ru|$). This is a genuine number-theoretic fact. However, 29 is **not** in the order of Co_0 (the Leech-lattice

automorphism group), is not a Niemeier Coxeter number, and is not the dimension of any standard Leech-related representation. The ratio $29/24 \approx 1.2083$ is not a clean Leech-rank / Coxeter ratio.

Post-hoc, one could read $39/29$ as a "cross-coupling" of D-Sink (13, internal to UBP) and Monster-prime (29, external): $G = (\text{Triad} \times \text{D-Sink} / \text{Monster-prime}) \times Y^{18} / \text{Wobble}$. This reading is poetic but unsupported — the search found $39/29$ because it is the unique rational that closes the gap to G ; the Monster-prime interpretation was attached afterwards. Per the prior study's own Critical Assessment point 4, this is exactly the kind of "post-fitting variable within the substrate ground" that the study warned against.

6. Quaternary Study: Q5 — Sextet Compound NRCI Decomposition

The prior study found a 6-clique of Golay octads at indices $[0, 2, 4, 6, 8, 10]$ — the "Sextet" — and reported its compound NRCI as 0.348, below the 0.70 Capture-Zone threshold. The study explicitly asked (Q5): what is the physical meaning of a compound NRCI below 0.70 for this geometric object?

We recomputed the per-octad symmetry tax and NRCI, then tested several aggregation rules. All six octads in the Sextet have identical tax = 3.1174 and NRCI = 0.7623 (the per-octad values reported in the prior study). The compound NRCI 0.348 arises from the standard UBP aggregation rule $NRCI = 10 / (10 + \sum tax_i)$ with the six taxes summed:

$$\text{compound_NRCI} = 10 / (10 + 6 \times 3.1174) = 10 / 28.7044 \approx 0.3484$$

To test whether this 0.348 value is special to the Sextet, we drew 200 random 6-octad samples from the 759 Golay octads and computed the compound NRCI for each. **All 200 samples produced NRCI = 0.3484 — identical to the Sextet.** The reason is structural: octads are weight-8 codewords of the binary Golay code, which form a single orbit under the Mathieu group M_{24} ; by symmetry, every octad has the same Leech-lattice symmetry tax. Therefore the compound NRCI of *any* 6 octads is 0.3484 — not just the Sextet.

Quantity	Value
Sextet per-octad tax	3.1174
Sextet per-octad NRCI	0.7623
Sextet compound NRCI (paper)	0.3480
Sextet compound NRCI (computed)	0.3484
Compound NRCI aggregation rule	sum-tax $\rightarrow$ $NRCI = 10 / (10 + \sum tax_i)$
Random 6-octad samples (n=200) — all give same NRCI?	YES (all 0.3484)
Sextet percentile in random distribution	0.0%

Conclusion for Q5. The Sextet's compound NRCI = 0.348 is not a special property of the Sextet — it is a generic property of any 6 octads, arising trivially from the standard aggregation rule applied to the (identical) per-octad taxes. The 0.70 Capture-Zone threshold is structurally unreachable for any 6-octad compound, because six taxes of 3.1174 always sum to 18.70, which always gives $NRCI = 0.348$. The "sub-threshold = subliminal / ground-state" interpretation proposed in the prior study therefore applies to *every* 6-octad set, not specifically to the Sextet. This is not necessarily wrong — it could be that the UBP framework intends 6-octad compounds as a class to be sub-threshold — but it does mean the Sextet is not distinguished by this property.

A separate observation: 0.348 is close to several substrate expressions ($w/2.35 = 0.3479$, $Y^2 \cdot 5 = 0.3503$, $Y \cdot 13/10 = 0.3441$) — but these are coincidences of magnitude, not derivations. The 0.348 value comes from the aggregation, not from any single substrate expression.

7. Critical Assessment

What this push achieves:

1. Q4 is partially answered, and the answer is mixed. The gravity-style grammar does *not* uniquely produce sub-0.13% hits for G. With the curated Y-power set $\{Y^{10}, Y^{12}, Y^{18}\}$, G is the only reachable target (Phase A). With the expanded Y-power set $\{Y^1..Y^{40}, Y_{inv}^1..Y_{inv}^{10}\}$, five targets (G, α , $\alpha^{\blacksquare}$, m_p/m_e , m_μ/m_e) are reachable with comparable accuracy (0.03-0.25%). The m_μ/m_e hit at 0.0294% is actually 3× better than the G hit. This is consistent with the substrate being broadly predictive, but also consistent with the grammar being broadly permissive.

2. The coincidence benchmark is unfavorable to the substrate. Across 29 700 candidates, the 0.13% band has 0-1 hits per target, but the 5% band has 28-54 hits. The grammar is permissive at lower accuracy. More importantly, the null-model test (40 scrambled-substrate trials) shows a 20-80% false-positive rate at the 0.13% threshold. The real substrate's G hit is at the 20th percentile of the null distribution — not at the extreme tail. For other constants, the real substrate *underperforms* the null model mean.

3. Q2 is partially resolved. 29 is a prime divisor of |Monster| — a real number-theoretic fact — but it has no clean Leech-lattice interpretation. The ratio 29/24 is most likely a search artifact, though one can post-hoc read it as a "Monster-prime / Leech-rank" coupling. This interpretation is poetic but unsupported.

4. Q5 is resolved, and the resolution is trivialising. The Sextet's compound NRCI = 0.348 is not a special property of the Sextet — any 6 octads give this value, because all Golay octads have the same symmetry tax (3.1174) by M_{24} symmetry. The "sub-threshold" interpretation therefore applies to *every* 6-octad compound, not specifically to the Sextet.

What this push does *not* achieve:

1. No falsification of UBP itself. The null-model result falsifies the *significance* of the gravity formula's 0.13% hit, but it does not falsify the UBP framework. UBP could still be correct as an ontology while the gravity formula is coincidence. To falsify UBP itself, one would need to show that *no* substrate-derived formula can predict any constant above the null-model false-positive rate — a much stronger test.

2. No closure of the 0.13% gap. Q1 (second-order correction) was not addressed in this push. The 0.00886×10^{11} gap between G_{UBP} and CODATA G remains. Given the null-model result, attempting to close this gap risks compounding the coincidence — each additional correction term increases the grammar's combinatorial freedom and therefore the false-positive rate. We recommend against chasing Q1 until Q4 is resolved more favourably.

3. No Q3 (Y^{18} derivability) test. The expanded search (Phase B) provides indirect evidence that Y^{18} is not unique (Y^{17} , Y^{15} also give sub-1% hits for G), but a proper Q3 test would require a structural derivation of Y^{18} from the UBP layer model, which is beyond the scope of this push.

Verdict:

The gravity formula's 0.1327% accuracy against CODATA is statistically indistinguishable from coincidence. Across 40 scrambled-substrate trials, 8 (20%) produced G hits of comparable quality. The substrate is broadly permissive — it produces sub-0.25% hits for 5 of 7 reachable targets — but this permissiveness is itself the most likely explanation for the G hit. The m_μ/m_e hit ($13/L = 169/w$, 0.0294%) is the only formula in this study with a signal-to-noise ratio worth follow-up. The Sextet's compound NRCI = 0.348 is a trivial consequence of M_{24} symmetry, not a special property. The 29/24 ratio is most likely a search artifact, though 29 is genuinely a Monster-prime. We do not claim UBP is false; we claim only that the gravity formula is not, on the present evidence, distinguishable from a permissive grammar's typical output.

8. Updated Open Questions

The prior study's five open questions are updated as follows. Items marked **[RESOLVED]** are now answered; items marked **[PARTIAL]** have new evidence but no consensus; items marked **[OPEN]** remain open.

ID	Status	Question	This push's contribution
Q1	[OPEN]	Can the 0.1327% gap between G_{UBP} and CODATA G be closed by a second-order correction?	Not addressed. We recommend against chasing this until Q4 is resolved, since adding correction terms increases the false-positive rate.
Q2	[PARTIAL]	Why 29/24 and 39/29? Independent geometric justification?	29 is a prime divisor of Monster (and Fi $\blacksquare\blacksquare\blacksquare$, Ru). 39 = 3·13 = Triad·D-Sink. But 29/24 has no clean Leech/Niemeier interpretation; likely a search artifact.
Q3	[PARTIAL]	Is Y^{18} truly predictive or post-hoc?	Phase B found Y^{17} (0.57%) and Y^{15} (0.65%) also give sub-1% G hits. Y^{18} is not unique; the layer interpretation is post-hoc.
Q4	[PARTIAL]	Does the structure appear for other physical constants?	Grammar hits 5/7 targets in sub-0.25%: G (0.13%), α (0.22%), $\alpha\blacksquare^1$ (0.22%), m_p/m_e (0.24%), m_μ/m_e (0.029%). α_G and $\Lambda\cdot\blacksquare_P^2$ are unreachable (dynamic range). But null model says 20-80% false-positive rate.
Q5	[RESOLVED]	Physical meaning of compound NRCI < 0.70 for the Sextet?	Trivial: any 6 octads give NRCI = 0.348, because all Golay octads have the same tax (3.1174) by $M\blacksquare\blacksquare$ symmetry. Sextet is not special.

Three new open questions emerge from this push:

NQ1. The m_μ/m_e formula $13/L = 169/w$ gives 0.0294% error — the best signal-to-noise ratio in this study. Does it survive a stricter null model (e.g., scrambling only Y , w , L while keeping the integer 13 fixed)? Is there an analogous formula for m_τ/m_e using $39 = 3\cdot 13$ or $169 = 13^2$?

NQ2. The grammar's permissiveness drops sharply between 1% (6-8 hits per target) and 0.13% (0-1 hits per target). Is there a structural reason for this "narrow spine"? Could the spine be derived from a subset of substrate constants (rather than the full grammar)?

NQ3. The Phase B Y -power picks (Y^3 for α , Y^{18} for G , Y_{inv}^6 for m_p/m_e , Y_{inv}^9 for m_τ/m_e) form a pattern: the layer-boundaries 6, 12, 18, 24 are not the picks, but 3, 6, 9, 18 (multiples of 3) are. Is there a UBP-internal reason why multiples of 3 should be the Y -power spectrum?

9. File Inventory

File	Type	Description
q4_generalisation_search.py	Script	Phase A — initial grammar replication (2 400 candidates per target, 7 targets)
q4_expanded_search.py	Script	Phase B — expanded Y-power sweep (29 700 candidates per target, 8 targets)
coincidence_null_model.py	Script	Phase C — coincidence spectrum + null model (40 scrambled-substrate trials)
q2_leech_29_39.py	Script	Q2 — Leech / Monster / Niemeier / sporadic-group enumeration for 29 and 39
q5_sextet_decomposition.py	Script	Q5 — Sextet per-octad tax + 200 random 6-octad null distribution
q4_generalisation.json	Data	Phase A results: top 5 per target, band counts
q4_expanded.json	Data	Phase B results: top 5 per target, Y-power picks, band counts
coincidence_null_model.json	Data	Phase C results: spectrum + null-model hit rates + percentile summaries
q2_leech_29_39.json	Data	Q2 results: Leech/Monster/Co $\blacksquare$ /Niemeier invariant table
q5_sextet.json	Data	Q5 results: per-octad tax, aggregation comparison, random-sample distribution
ubp_unified_v5.py	Core	v5.3 hardened triad-physics edition, float-free core (unchanged, copied from prior session)

All scripts persist in /home/z/my-project/scripts/; all result data in /home/z/my-project/results/. All numerical computations use Python fractions.Fraction exact rational arithmetic via the v5.3 ExactMath / ExactRoot subsystem; no floating-point arithmetic was used inside the computational core. Floats appear only at the display boundary for legibility.

Appendix A. Sporadic Group Orders and 29-Divisibility

Reference table for Q2 — the 26 sporadic groups, their standard names, and whether their order is divisible by 29. The Monster group M, the Fisher group Fi_{24}' , and the Rudvalis group Ru all contain 29 as a prime factor of their order. No other sporadic group does. The Conway groups Co_0 , Co_1 , Co_2 , Co_3 (the Leech-lattice automorphism family) do **not** contain 29.

Group	Order (exact)	Factorisation	29 divides?
M11	7,920	$2^4 \cdot 3^2 \cdot 5 \cdot 11$	no
M12	95,040	$2^4 \cdot 3^3 \cdot 5 \cdot 11$	no
M22	443,520	$2^4 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$	no
M23	10,200,960	$2^4 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 23$	no
M24	244,823,040	$2^4 \cdot 3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23$	no
J1	175,560	$2^3 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 19$	no
J2	604,800	$2^4 \cdot 3^3 \cdot 5^2 \cdot 7$	no
J3	50,232,960	$2^4 \cdot 3^4 \cdot 5 \cdot 17 \cdot 19$	no
J4	8.6776e+20	$2^{21} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11^2 \cdot 23 \cdot 29 \cdot 31 \cdot 37 \cdot 43$	YES

Group	Order (exact)	Factorisation	29 divides?
Co1	4.1578e+18	$2^{2^1} \cdot 3 \cdot 5 \cdot 7^2 \cdot 11 \cdot 13 \cdot 23$	no
Co2	42,305,421,312,000	$2^1 \cdot 3 \cdot 5^3 \cdot 7 \cdot 11 \cdot 23$	no
Co3	495,766,656,000	$2^1 \cdot 3 \cdot 5^3 \cdot 7 \cdot 11 \cdot 23$	no
Fi22	64,561,751,654,400	$2^1 \cdot 3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13$	no
Fi23	4.0895e+18	$2^1 \cdot 3^1 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 23$	no
Fi24'	1.2552e+24	$2^{2^1} \cdot 3^1 \cdot 5^2 \cdot 7^3 \cdot 11 \cdot 13 \cdot 17 \cdot 23 \cdot 29$	YES
HS	44,352,000	$2 \cdot 3^2 \cdot 5^3 \cdot 7 \cdot 11$	no
McL	898,128,000	$2 \cdot 3 \cdot 5^3 \cdot 7 \cdot 11$	no
He	4,030,387,200	$2^1 \cdot 3^3 \cdot 5^2 \cdot 7^3 \cdot 17$	no
Ru	145,926,144,000	$2^1 \cdot 3^3 \cdot 5^3 \cdot 7 \cdot 13 \cdot 29$	YES
Suz	448,345,497,600	$2^{1^3} \cdot 3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13$	no
O'N	460,815,505,920	$2 \cdot 3 \cdot 5 \cdot 7^3 \cdot 11 \cdot 19 \cdot 31$	no
HN	273,030,912,000,000	$2^1 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 19$	no
Ly	5.1765e+16	$2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 31 \cdot 37 \cdot 67$	no
Th	9.0746e+16	$2^1 \cdot 3^1 \cdot 5^3 \cdot 7^2 \cdot 13 \cdot 19 \cdot 31$	no
B	4.1548e+33	$2^{1^1} \cdot 3^{1^3} \cdot 5 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 31 \cdot 47$	no
M	8.0802e+53	$2 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11^2 \cdot 13^3 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71$	YES

Observation: 29 is a relatively rare prime in sporadic-group orders — only 3 of 26 sporadics (M, Fi_{24}' , Ru) contain it. The fact that 29 appears in the gravity formula's 39/29 ratio could therefore be read as a coupling to the Monster family. But this is post-hoc: the search found 29 because 39/29 is the rational that closes the G gap, not because of any structural link to the Monster. A truly structural derivation would need to show why a Monster-prime (rather than, say, a Leech-rank divisor) should appear in the gravitational coupling.

Appendix B. Per-Octad Symmetry Tax (Q5 supporting data)

Full per-octad decomposition of the Sextet (indices [0, 2, 4, 6, 8, 10]) and the standard aggregation rules applied. All six octads have identical symmetry tax = 3.1174 and NRCI = 0.7623 — a consequence of the M_{24} -transitivity of weight-8 Golay codewords. The compound NRCI of 0.3484 arises from the standard UBP aggregation $\text{NRCI} = 10 / (10 + \Sigma \text{tax}_i)$ and is therefore the same for *any* 6-octad set.

Octad index	Symmetry tax	Per-octad NRCI	Within Capture Zone?
0	3.1174	0.7623	YES (≥ 0.70)
2	3.1174	0.7623	YES (≥ 0.70)
4	3.1174	0.7623	YES (≥ 0.70)
6	3.1174	0.7623	YES (≥ 0.70)
8	3.1174	0.7623	YES (≥ 0.70)
10	3.1174	0.7623	YES (≥ 0.70)

Aggregation rule	Formula	Compound NRCI	Matches paper (0.348)?
Sum tax → NRCI (standard UBP)	$10 / (10 + \sum \text{tax}_{\blacksquare})$	0.3484	YES
Arithmetic mean of NRCIs	$\sum \text{NRCI}_{\blacksquare} / 6$	0.7623	no
Geometric mean of NRCIs	$(\prod \text{NRCI}_{\blacksquare})^{(1/6)}$	0.7623	no
Harmonic mean of NRCIs	$6 / \sum (1/\text{NRCI}_{\blacksquare})$	0.7623	no
Product of NRCIs	$\prod \text{NRCI}_{\blacksquare}$	0.1963	no
NRCI from mean tax	$10 / (10 + \text{mean}(\text{tax}_{\blacksquare}))$	0.7623	no

The standard UBP rule (sum tax → NRCI) is the only aggregation that matches the paper's 0.348 value. This is by design — the UBP framework defines compound NRCI this way. The other aggregation rules give 0.762 (mean, harmonic, geometric) or 0.196 (product), neither of which is sub-threshold. The compound-NRCI sub-threshold property is therefore **not** an intrinsic property of the Sextet's geometry; it is a consequence of the specific aggregation rule chosen by UBP. A different (equally defensible) aggregation rule would keep the Sextet in the Capture Zone.

Appendix C. Null Model Error Distributions

Per-target distribution of best-error-percent across 40 scrambled-substrate trials. The real substrate's best error is compared to the null-model distribution. If the real best is in the lower tail (e.g. below the 10th percentile of the null distribution), that is evidence the real substrate is genuinely better than random. If the real best is at or above the null-model median, the real substrate is statistically indistinguishable from random.

Target	Real best %	Null min %	Null p10 %	Null p50 %	Null p90 %	Null max %	Real > p50?
G	0.1327	0.0229	0.0577	0.4150	2.1897	2.6094	no
alpha	0.2219	0.0019	0.0156	0.1317	0.7297	1.2511	YES (real is worse)
alpha_inv	0.2223	0.0002	0.0092	0.0670	0.3619	0.7941	YES (real is worse)
mp/me	0.2427	0.0045	0.0223	0.1033	0.3803	0.7570	YES (real is worse)
mmu/me	0.0294	0.0036	0.0129	0.0467	0.2462	0.7167	no
mtau/me	0.4929	0.0021	0.0094	0.0669	0.3537	0.6446	YES (real is worse)
alpha_G	3636590468 56637504.0 000	157430064 454922.59 38	394127955 40592072. 0000	806206012 29949168. 0000	276943489 874951104 .0000	100792567 775857228 8.0000	YES (real is worse)

Reading: for G, the real best (0.1327%) is below the null-model median (0.50% approx) but well above the null-model minimum (0.023%) — i.e., the real substrate is in the lower half of the null distribution but not at the extreme tail. For α , $\alpha_{\blacksquare}^{-1}$, m_p/m_e , m_μ/m_e , m_τ/m_e , the real best is **above** the null-model median, meaning the real substrate is *worse* than the typical scrambled substrate. This is the strongest single piece of evidence in this study that the substrate constants are not specially tuned for predicting these physical constants.

A caveat: the null model is conservative. It scrambles each substrate constant independently by up to two orders of magnitude, but it preserves the grammar structure (bases, scales, multipliers) and the rough magnitude of each constant. A more aggressive null model — one that also scrambles the multipliers and the choice of which constants to include as bases — would likely produce even higher false-positive rates. The 20-80% false-positive rates reported here are therefore lower bounds on the true false-positive rate.

Appendix D. Methodology Detail — Null Model Construction

This appendix documents the precise construction of the null model used in §4.2, so that the test can be reproduced and critiqued. The design choices below each affect the false-positive rate; we discuss each and explain why we chose the conservative variant.

D.1 Substrate scrambling rule

For each trial, each substrate constant c_i (drawn from $\{Y, Y_{inv}, L, L_s, \pi, \phi, e, w, U_e, NRCI\}$) is replaced by $c_i' = c_i \times u_i$, where u_i is drawn independently from $\text{uniform}(0.1, 10.0)$. The result is then converted to a Fraction with denominator limited to 10^6 for tractability. This scrambling preserves the order of magnitude (within ± 1 dex) of each constant, which is the minimum needed to keep the grammar's dynamic range comparable.

Why not scramble more aggressively? Scrambling $c_i' = \text{uniform}(0, 1)$ (a fully random Fraction in $[0,1]$) would destroy the magnitude structure of the substrate and produce a null model that mostly fails to hit any target. That would be a straw-man test. The conservative variant (within 1 dex) gives the null model a fair chance while still removing any structural significance from the precise values of the constants.

Why not scramble less aggressively? Scrambling $c_i' = c_i \times \text{uniform}(0.5, 2.0)$ (within ± 0.3 dex) would preserve most of the substrate's structure and would produce a less informative null model. The 1-dex window is a reasonable compromise: large enough to destroy precise structural information, small enough to preserve dynamic range.

D.2 Grammar preservation

The grammar (10 bases $\times$ 2 directions $\times$ 55 scales $\times$ 27 multipliers = 29 700 candidates) is held fixed across all trials. Only the substrate constant values are scrambled. This is conservative: scrambling the grammar too (e.g. random multipliers, random subset of bases) would increase the false-positive rate. The fixed grammar gives the real substrate its best chance to outperform the null model.

One subtlety: the Y^k scale family uses the scrambled Y value. So if Y is scrambled to $Y' = 10 \cdot Y$, then $Y'^{18} = 10^{18} \cdot Y^{18}$, which is a very different scale. This is intentional — it tests whether the precise value of Y (not just its role as a Y -power base) is what makes the G hit work. The fact that 20% of scrambled substrates still hit G within 0.13% means the precise Y value is **not** load-bearing for the hit — many other Y values would work too.

D.3 Aggregation across trials

For each target and each trial we record (a) the best error across all 29 700 candidates, (b) whether the best error is $\leq 0.13\%$, (c) whether it is $\leq 0.05\%$. Across 40 trials we then report the count of $\leq 0.13\%$ hits and the distribution of best errors. The 40-trial sample size gives a 95% confidence interval of approximately ± 15 percentage points on the false-positive rate (binomial). So a reported 20% false-positive rate has a 95% CI of roughly $[8\%, 35\%]$ — still well above 0, and still incompatible with the real substrate being statistically surprising.

Why 40 trials and not 1000? Each trial evaluates $29\,700 \times 7 = 207\,900$ candidate formulas, each requiring exact-rational arithmetic on Fractions with up to 10^6 denominators. 40 trials complete in approximately 4 minutes; 1000 trials would take ~ 100 minutes. The 40-trial result is already decisive: even with the 95% CI, the false-positive rate is nowhere near 0%.

D.4 What would change the verdict?

The null-model verdict ("the 0.13% G hit is statistically indistinguishable from coincidence") would be reversed if any of the following held:

(a) The real substrate's G hit rate were in the extreme tail of the null distribution — e.g. if the real 0.1327% were below the null-model minimum (0.0229%). It is not: the null-model minimum is $6\times$ smaller than the real best.

(b) The real substrate's *structural* property of the formula (e.g. the specific Y -power chosen) were not reproducible by scrambled substrates. But the null model produces Y -power picks across the same range $\{3, 6, 9, 18\}$ as the real substrate, so this structural property is also reproducible by chance.

(c) The formula had an out-of-sample prediction — e.g. if it predicted a constant that was not used in the search's design (like $\Lambda \cdot \mu_p^2$). It does not: the formula fails catastrophically for $\Lambda \cdot \mu_p^2$ (off by $10^3\%$).

Since none of (a), (b), or (c) holds, the verdict stands. We note, however, that the m_μ/m_e formula $13/L = 169/w$ *does* satisfy a weaker version of (a): its 0.0294% error is below the null-model 50th percentile (0.10%) by a factor of 3.4. This is the only formula in the study that approaches a meaningful signal-to-noise ratio.

Appendix E. Recommended Next Steps

Given the push's findings, we recommend three concrete next steps ordered by expected information gain.

E.1 Deepen the m_μ/m_e investigation

The formula $13/L = 169/w$ predicts m_μ/m_e with 0.0294% error — the best signal-to-noise ratio in this study. Crucially, it uses no Y-power and no integer close to the target value (206.77). It is structurally simpler than the G formula. We recommend:

- (i) Run the null-model benchmark on this formula specifically: how often does $13/L'$ (with $L' = w'/13$ and w' a random fraction) hit 0.0294% on m_μ/m_e ? If the false-positive rate is low (e.g. $<5\%$), this formula is genuinely surprising.
- (ii) Search for analogous formulas: does $39/L = 507/w$ predict m_μ/m_e ? Does $169/L = 13^2/L$ predict m_μ/m_e ? The pattern " $D\text{-Sink}^k / L$ " would be a strong UBP-internal prediction if it generalises across lepton generations.
- (iii) Cross-check against the PARTICLE_PHYSICS atlas: the existing atlas uses $m_\mu/m_e = 206 + 12 \cdot L = 206.755$ (0.0066% error, lens = Core Ratio). The new formula $13/L = 206.761$ (0.0294% error) is less accurate but structurally cleaner — no integer close to 206. Which is more likely to be the "true" substrate formula?

E.2 Build a structural null model

The current null model scrambles constant values but preserves grammar structure. A stronger test would scramble the grammar too: random subsets of bases, random multipliers, random Y-power ranges. If the real substrate's hit rate is still above this stronger null, that would be stronger evidence of structural significance. If not, the permissiveness is in the grammar, not the substrate.

Concretely: generate 100 random grammars by sampling 5-15 bases from a pool of 30 candidate constants (substrate constants + random Fractions), 10-30 random multipliers from a pool of 50, and 5-20 random scales. Run each grammar on the real substrate and on scrambled substrates. The real substrate's hit rate should be compared to the distribution of scrambled-substrate hit rates across random grammars. This is a two-dimensional null model (grammar $\times$ substrate) and is the strongest falsification test we can devise without introducing new physics.

E.3 Test out-of-sample predictions

The most decisive test of any physical formula is out-of-sample prediction. The current study tested constants whose values are well-known and could have influenced the search design (even unintentionally). A truly out-of-sample test would be:

- (i) A constant not yet measured precisely — e.g. the Hubble constant H_0 (current tension between CMB and supernovae values). Does the substrate predict a specific value in the contested range?
- (ii) A ratio not yet targeted by UBP — e.g. the W/Z boson mass ratio (currently measured at $80.379 / 91.1876 = 0.8826$). Does the substrate hit this without it being in the search design?
- (iii) A constant with a predicted time-variation — e.g. if α is substrate-determined, does the substrate predict a specific drift rate for α ? (Current observational bound: $|d\alpha/dt|/\alpha < 10^{-17}/\text{yr.}$)

Any of these would be a stronger test than the in-sample fits we have performed. We recommend the user consider (i) and (ii) as the next round of UBP gravity-substrate validation.